Employee Management System
BugAnalysisReport

Bug Number	Error Description	Root Cause	Resolution
1	Type mismatch on getEmployeeById parameter	The function expected a string, but Employee.id is a number.	Changed parameter type to id: number.
2	Potential undefined return type error	.find() can return undefined, but method signature only promised Employee.	Updated return type to Employee | undefined.
3	Object is possibly 'undefined' on calculateBonus	salary is an optional property ? in the interface, causing math operations to fail if undefined.	Added a ternary fallback check (employee.salary ? ... : 0).
4	toUppercase is not a function	Typo in the JavaScript standard library string method name.	Corrected to toUpperCase().
5	Type 'string' is not assignable to type 'number'	Added employee ID "102" was passed as a string.	Removed quotes to pass 102 as a number.
6	Object is possibly 'undefined' during execution	Calling employee.department without verifying if getEmployeeById successfully found a record.	Wrapped the execution calls in an if (employee) block to ensure type safety.